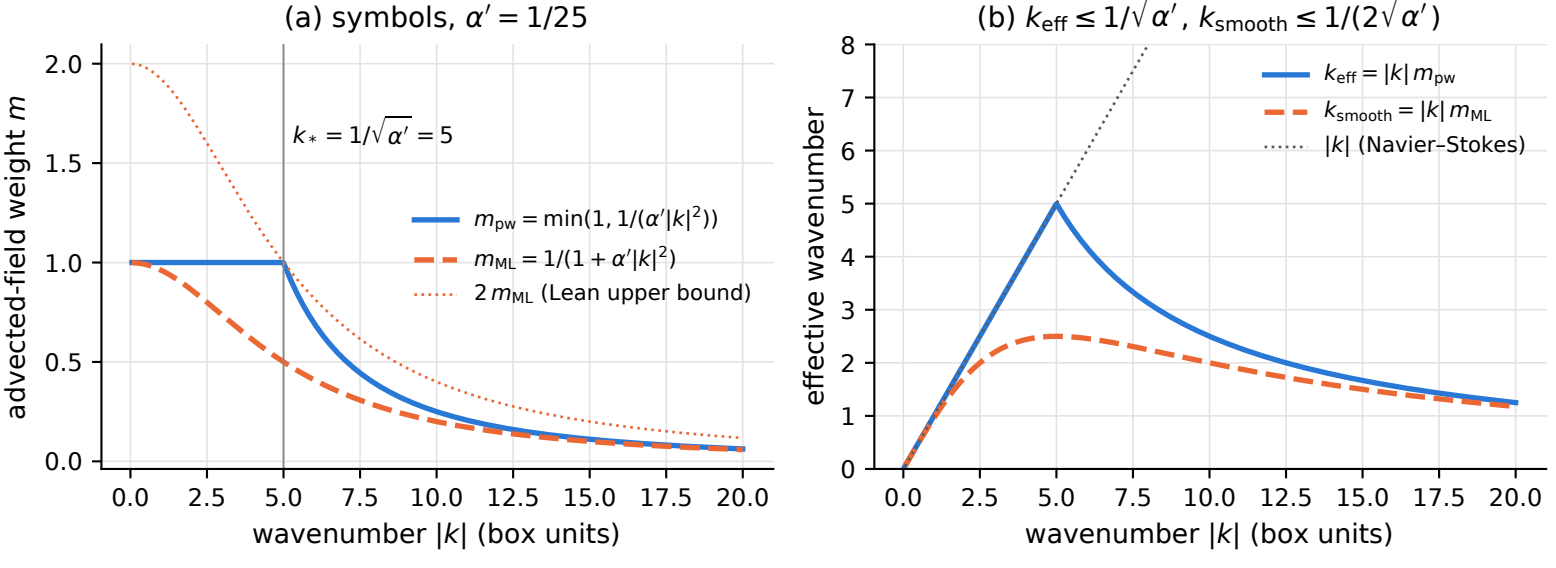
(a) symbols, $\alpha' = 1/25$
advected-field weight m
$k_* = 1/\sqrt{\alpha'} = 5$
$m_{\mathrm{pw}} = \min(1, 1/(\alpha'|k|^2))$
$m_{\mathrm{ML}} = 1/(1+\alpha'|k|^2)$
$2\,m_{\mathrm{ML}}$ (Lean upper bound)
wavenumber $|k|$ (box units)
(b) $k_{\mathrm{eff}} \le 1/\sqrt{\alpha'}$, $k_{\mathrm{smooth}} \le 1/(2\sqrt{\alpha'})$
effective wavenumber
$k_{\mathrm{eff}} = |k|\,m_{\mathrm{pw}}$
$k_{\mathrm{smooth}} = |k|\,m_{\mathrm{ML}}$
$|k|$ (Navier–Stokes)
wavenumber $|k|$ (box units)